



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

CyberSecurity

1 - “Introduction to Computer Security”

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**This block of slides does not refer
to a specific chapter in the book.**

It is a “gentle introduction”

CyberSecurity

1 - “Introduction to Computer Security”

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A few **basic principles** of computer security:

- principle 1: "*non-existence of secure systems*"
it is a logical and immediate consequence of a principle that has already been demonstrated.
- principle 1 (**corollary**): "*systems complexity*"
- principle 2: "*entities composing a system*"
- principle 3: "*security = knowledge*"

Security of a **cryptosystem** (i.e., *security through obscurity*)

Principle 1: on “Secure Systems”



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SECURE SYSTEMS DO NOT EXIST

- **The software is often far from perfect.** It is full of **implementation bugs** and **design errors.** It is the same for **communication protocols**
- **The “unbreakable system” is only a myth.** Just like the impenetrable bank vault or the unsinkable boat (e.g., *RMS Titanic*)

Principle 1: on “Secure Systems”



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SECURE SYSTEMS DO NOT EXIST

An attacker will attack a system if there is enough value to make the attack worth it

Security is costly: so there are various level of security.
It is very difficult to accurately assess the level of security required for a system.

The **security level** of a given system is determined by the **amount of**

(amount of time it takes an attacker to find a vulnerability, develop an exploit, and breach the system)

(How much does it cost an attacker to breach the system?)

time necessary to break the system, the **amount of money** (or

necessary to break the system

resources) and the **probability of success**

of the attack

Vulnerabilities can be discovered (by a paid individual or a criminal)

Drastically reducing the probability of success (e.g., by implementing two-factor authentication) discourages attackers, because they know they will have to make a huge number of attempts (increasing both time and costs) before succeeding.

Principle 1: on “Secure Systems”



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How much am I willing to invest in identifying and fixing vulnerabilities in my system?

SECURE SYSTEMS DO NOT EXIST

What Are We Protecting? (The Assets)
- Hardware & Infrastructure: Servers, network devices, and cloud computing resources.
- Software & Services: Source code, applications, operating systems, and configurations.
- Data & Information: User credentials, financial records, intellectual property, and personal data (e.g., GDPR-regulated info).

- The **security level** of a given system is determined by the **amount of time** necessary to break the system, the **amount of money** (or resources) and the **probability of success**

What are we protecting?

Principle 1: on “Secure Systems”



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SECURE SYSTEMS DO NOT EXIST

From Whom? (Threat Actors)

- Script Kiddies: Amateurs using pre-made tools for fun or notoriety.
- Cybercriminals: Well-organized professionals driven purely by financial gain.
- Insider Threats: Disgruntled, malicious, or negligent employees and partners.
- Hacktivists: Groups attacking systems to promote a political or ideological agenda.
- State-Sponsored Actors (APTs): Highly funded nation-state hackers focused on long-term espionage and sabotage.

- The **security level** of a given system is determined by the **amount of time** necessary to break the system, the **amount of money** (or resources) and the **probability of success**

Understanding who the enemy is, is important for determining the appropriate level of security.

What are we protecting? From **who**?

Principle 1: on “Secure Systems”



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SECURE SYSTEMS DO NOT EXIST

Why? (The Reasons for an Attack)

- Financial Gain (Money): Ransomware extortion, direct theft of funds, or unauthorized crypto-mining (Cryptojacking).
- Espionage: Stealing trade secrets, patents, or state intelligence for a strategic/competitive advantage.
- Sabotage & Warfare: interruption of critical infrastructure (power grids, transit) or destroying a competitor's capability.
- Ideology & Ego: Damaging a target's reputation (Defacement) or gaining prestige within hacker communities.

- The **security level** of a given system is determined by the **amount of time** necessary to break the system, the **amount of money** (or resources) and the **probability of success**

What are the reasons for a possible attack?

What are we protecting? From **who?** **Why?**

Principle 1: on “Secure Systems”



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Think in terms of value because that is the right way to assess the system's security level

SECURITY EXIST

- The **security level** is the **amount of** **time** necessary to compromise the system (or resources) and the **cost** (or **money**) required to do so
- Most times the ^{attacker} adversary is rational
- Most times the adversary has an economic goal (that is... \$\$\$)
- This is not always true when the adversary is a government

What are we protecting? From who? Why?

The Proactive Approach (Threat Modeling): Take action before a security incident occurs. You put yourself in the attacker's shoes while the system is still in the design phase to identify potential vulnerabilities and fix them before writing the final code.



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Principle 1: on “Secure Systems”

- This is an “informal way” for defining a

THREAT MODEL

By defining the threat model, I can determine the level of security the system requires

↑ creates

- Threat modeling is a structured process
used in cybersecurity to identify and
analyze potential threats and
vulnerabilities in a system or application

DO NOT EXIST

The Core Goals of Threat Modeling are:

- Understand the system's architecture and attack surface.
- Identify what assets are targeted.
- Identify and analyze potential threats and vulnerabilities.
- Define specific countermeasures before code deployment.

determined by the **amount of**

the **amount of money** (or

success

What are we protecting? From who? Why?

ZERODIUM Payouts for Mobiles*

Price tags

Up to \$2,500,000

Up to \$2,000,000

Up to \$1,500,000

Up to \$1,000,000

Up to \$500,000

Up to \$200,000

Up to \$100,000

Vulnerability Broker collects and sells vulnerabilities discovered by researchers (entities that buy and sell information about software vulnerabilities—usually zero-days—from security researchers, acting as intermediaries.)

chain of vulnerabilities to gain root privileges (e.g., vulnerabilities in WhatsApp and the kernel)

FCP: Full Chain with Persistence
RCE: Remote Code Execution
LPE: Local Privilege Escalation
SBX: Sandbox Escape or Bypass

■ iOS
■ Android
■ Any OS

4 types of exploit

When the device is restarted, the system remains compromised.

Zerodium purchases undisclosed cybersecurity vulnerabilities (known as zero-days) from independent researchers and hackers around the world. Zerodium then resells this information and the associated attack methods (exploits) to its select clients, which are primarily government agencies, intelligence services, and law enforcement agencies (for espionage or cyberwarfare purposes).

The price paid by Zerodium to purchase the vulnerability from researchers

1.001
Android FCP
Zero Click
Android

1.002
iOS FCP
Zero Click
iOS

code that runs automatically without user interaction

2.001
WhatsApp
RCE+LPE
Zero Click
iOS/Android

2.002
iMessage
RCE+LPE
Zero Click
iOS

The chart shows how much money Zerodium is willing to pay for a single working exploit targeting mobile devices (Android and iOS). On the vertical axis, we find the severity of the attack:

- RCE (Remote Code Execution): it allows the attacker to execute malicious code on the victim's phone remotely, without having physical access to the device.
- LPE (Local Privilege Escalation): the attacker already has limited initial access to the device and exploits this vulnerability to obtain maximum system privileges (e.g., Root or Administrator permissions).
- SBX (Sandbox Escape): allows the attacker to break out of the "security sandbox" in which apps run in isolation to access data from other applications.
- FCP (Full Chain with Persistence): a combined attack that merges RCE, SBX, and LPE, allowing the phone to be completely and permanently compromised.

2.003
WhatsApp
RCE+LPE
iOS/Android

2.004
SMS/MMS
RCE+LPE
iOS/Android

3.001
Persistence
iOS

2.005
WeChat
RCE+LPE
iOS/Android

2.006
iMessage
RCE+LPE
iOS

2.007
FB Messenger
RCE+LPE
iOS/Android

2.008
Signal
RCE+LPE
iOS/Android

2.009
Telegram
RCE+LPE
iOS/Android

2.010
Email App
RCE+LPE
iOS/Android

4.001
Chrome
RCE+LPE
Android

4.002
Safari
RCE+LPE
iOS

5.001
Baseband
RCE+LPE
iOS/Android

6.001
LPE to
Kernel/Root
iOS/Android

2.011
Media Files
RCE+LPE
iOS/Android

2.012
Documents
RCE+LPE
iOS/Android

4.003
SBX
for Chrome
Android

4.004
Chrome RCE
w/o SBX
Android

4.005
SBX
for Safari
iOS

4.006
Safari RCE
w/o SBX
iOS

7.001
Code Signing
Bypass
iOS/Android

5.002
WiFi
RCE
iOS/Android

5.003
RCE
via MitM
iOS/Android

6.002
LPE to
System
Android

8.001
Information
Disclosure
iOS/Android

8.002
[k]ASLR
Bypass
iOS/Android

9.001
PIN
Bypass
Android

9.002
Passcode
Bypass
iOS

9.003
Touch ID
Bypass
iOS

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ZERODIUM Payouts for Mobiles*

Up to
\$2,500,000

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Up to
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Up to
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Up to
\$500,000

Up to
\$200,000

Up to
\$100,000

FCP: Full Chain with Persistence
RCE: Remote Code Execution
LPE: Local Privilege Escalation
SBX: Sandbox Escape or Bypass

■ iOS
■ Android
■ Any OS

While RCE allows attackers to execute commands from a remote location to a device, LPE allows a low-privileged user to gain administrative or root-level access.

A full-chain vulnerability on Mobile is a sophisticated cyberattack that links together multiple individual software flaws, often zero-day vulnerabilities, to achieve complete, unauthorized control over a device. A full chain links, for example, a browser exploit, a sandbox escape, and a kernel privilege escalation to move from an external attack to full "root" or kernel-level control.

Is not a full-chain vulnerability

WhatsApp RCE example:

CVE-2022-36934

Exploit Whatsapp to be able to run unauthorized code on the device and then doing something on the device itself (maybe there is something after that maybe another privilege escalation)



Idea:
Exploit a Vulnerability in the Application and then exploit a Kernel Vulnerability to take control of Root Privileges



Zerodium ✓ @Zerodium · Jun 1

We're looking for #0day exploits affecting Pidgin on Windows and Linux.

Bounty: \$100,000

Read more: zerodium.com/temporary.html



9



129



213



Gary Kramlich

@rw_grim

Most of the time, the attacker is rational

The developer can introduce vulnerabilities and selling them

A lot of software products rely on open-source libraries

Replying to @Zerodium

This really shows the sad state of funding in open source software.. I managed to find 25k USD last year to work on @impidgin full time, but if you can find a security exploit in my and other's non paid work you can make 4 times as much as I did... 🤔

9:59 PM · Jun 1, 2021 · Twitter Web App

Idea Cloud Computing:

Exploit App Vulnerability, Exploit Virtual OS
Kernel Vulnerability and Exploit Virtualization
Software Vulnerability (to escape the VM) to
arrive on the HW or OS of the Machine

Vulnerability Brokers are against the Fix of the
Vulnerability (the Manufacture wants to fix it)

So, a researcher that has found a vulnerability
can:

- 1) Sell the Vuln to Manufacture (Participate to Bug
Bounty Program: get way less money than from
Vuln. Brokers)
- 2) Sell the Vuln to Vuln. Broker

(COROLLARY)

Principle 1: on “Complexity of Systems”



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MORE COMPLEXITY → LESS SECURITY

As a system grows in size and functionality, maintaining a high level of security becomes exponentially harder.

Why Security Fails in Complex Systems:

- Larger Attack Surface: More features mean more potential entry points for attackers.
- The Human Limit: No single engineer can understand the entire codebase and its dependencies.

● (Empirical rule) Increasing the complexity of a systems leads to an insecure system

→ Vulnerabilities rarely emerge from a single isolated component; rather, they arise from complex and unexpected interactions between multiple secure components.

● Linear increase of complexity means super-linear increase of insecurity

Principle 1: on “Complexity of Systems”



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MORE COMPLEXITY → LESS SECURITY

- Do you want to build a “**reasonably secure**” system?

Reduce the Complexity
of the System

KISS RULE:

- avoid unnecessary complications.

Keep It Simple and Stupid

- Keep It Simple, Stupid*
- Keep It Short and Stupid*

If you don't understand the
whole system, it is very difficult to
have that system secure

Principle 1: on “Complexity of Systems”



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MORE COMPLEXITY → LESS SECURITY

- Does this mean that the systems to be secured can **not** be **complex** or **complicated**?

Principle 1: on “Complexity of Systems”



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MORE COMPLEXITY → LESS SECURITY

- Does this mean that the systems to be secured can **not** be **complex** or **complicated**?
 - High complexity is inevitable in modern systems.
 - While the underlying system might be complex, the protection system must remain as simple as possible.

NO!



Principle 1: on “Complexity of Systems”

Example:

- Complex Scenario (Wrong): To secure access to servers, you create a custom proprietary system with 5 different encryption steps, tokens that expire every 12 seconds, which requires three different databases to verify identity, and which changes rules depending on the time of day. Result: A nightmare to manage. A minor bug in any of these steps will block legitimate users or create a huge logical flaw that a hacker will exploit.

- Simple Scenario (KISS - Correct): To secure the exact same complex server, use a standard, open, and widely tested protocol like SSH with public/private cryptographic keys and simple two-factor authentication (2FA). Result: The mechanism is straightforward, clean, very easy to verify, and leaves very little room for human error.

MORE COMPLEXITY → LESS SECURITY

It is the most important part of the whole system (example: Authentication of the device to use it)

The **protection system** must be “*as simple as possible*”. If possible...
“*very simple*”

- **Unnecessary complexity** ^{in security systems} can lead to {**design, implementation, usage**} errors
- Each **error** is a potential “**vulnerability**”

Every added complexity -> Potential Error -> Exploitable Vulnerability.

Principle 2: on “Entities composing a system”



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These are the zones of the system where we can have a Vulnerability. A system consists of three interconnected components, and its overall security depends on the weakest link in the chain.

(MOST) SYSTEMS ARE COMPOSED OF 3 TYPES OF ENTITIES

- Software
- Hardware
- Humanware Humanware represents the people who interact with the system: network administrators, developers, managers, and customers.

Principle 2: on “Entities composing a system”



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The attack surface is the set of all points where an unauthorized user (the attacker) can attempt to insert data, extract data, or interact with the system to find a vulnerability. Modern software has a massive attack surface because it consists of:

- User interfaces and APIs.
- Open network ports and protocols: services listening to exchange data (HTTP, FTP, SSH).
- Third-party code (dependencies): Open-source libraries that developers import so they don't have to rewrite everything from scratch. If one of these external libraries has a vulnerability (as happened with the famous Log4Shell case), the entire application becomes vulnerable, even if the code you wrote is flawless.

SOFTWARE

- Example of a software attack (Difficult): Spend weeks analyzing the binary code of a banking application, find a memory bug (buffer overflow), write a complex exploit to bypass the operating system's defenses, and inject malicious code. This requires extraordinary skills and a great deal of time.
- Example of an alternative attack (Easy): Send a targeted phishing email to the bank's administrator (Humanware attack) pretending to be the director requesting an urgent check, prompting them to enter their credentials on a clone site. Within 10 minutes, the hacker is inside with full privileges, without having touched a single line of the banking software's code.



It is a critical part of the system with a **large exposed attack**

surface

- remember that... **often other parts of the system** are “**easier to**

(hardware or humanware)

attack”





- Many, very complex programs
- Each program made of thousands or millions of lines of source code
(i.e. computer instructions)

- It is a critical part of the system with a **large exposed attack surface**
- remember that... **often other parts of the system are “easier to attack”**

Principle 2: on “Entities composing a system”



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SOFTWARE

Air-Gapped Computer: This refers to a computer system that is physically isolated from all insecure networks, including the internet. The name implies there is a "gap of air" between the machine and any network, offering maximum security against remote, network-based, or malware attacks.

- It is a critical part of the system with a **large exposed attack**

S

That can be **accessed remotely** (i.e. from the Internet, from a wireless network or from a user interface)

(example: cars)

the system are “**easier to**

attack

Principle 2: on “Entities composing a system”



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SOFTWARE

doing a well attack surface
analysis isn't so easy

- It is a critical part of the system with a **large exposed attack**

surface

tech or not (example: a paper document containing information about a company is considered a significant part of an organization's physical attack surface)

- remember ***Everything*** that can be used by an system are “**easier to**
attack” **attacker to violate the system**

Principle 2: on “Entities composing a system”



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SOFTWARE

- It is a critical part of the system with a **large exposed attack surface**
- remember that... **often other parts of the system are “easier to attack”**

Principle 2: on “Entities composing a system”



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Attacking the software often
requires some technical skills...

attacking the human in the
system requires skills that are not

about computing (psychological skills.
Example of attack: phishing)

SOFTWARE

system with a large exposed attack

- remember that... often other parts of the system are “easier to attack”

Principle 2: on “Entities composing a system”



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When it is not possible to update the firmware to fix vulnerabilities, organizations must shift from applying patches to implementing mitigation measures, such as removing the hardware or reducing its attack surface, to manage the risks. Firmware vulnerabilities have a significant impact because they operate below the operating system level, often allowing persistent and invisible access.

HARDWARE



• All the modern hardware is composed of **hardware AND software**

(that is called **firmware**)

Firmware is a specific type of software that is built directly into a hardware component. Firmware can contain bugs and vulnerabilities just like standard software.

• The hardware can be **tampered** by an attacker

Examples: Inserting hardware keyloggers, stealing unencrypted hard drives, or exploiting supply chain vulnerabilities by altering components before delivery.

• Even the **physical security** of systems and devices (**e.g. to deny unauthorized access**) is hard to guarantee

If an attacker has unrestricted physical access to a device, it can no longer be trusted. Why it is hard to guarantee: Modern infrastructure is decentralized, making physical perimeters insufficient to protect all assets.

Principle 2: on “Entities composing a system”



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HARDWARE

- All the modern hardware is composed of **hardware AND software** (that is called **firmware**)

- This means that an attacker can modify the hardware or ^{compromise} ~~act on~~ the software (e.g. flashing
- a modified firmware) if the firmware is updatable and not well secured, the attacker can install a firmware with a backdoor

Principle 2: on “Entities composing a system”



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To interfere with a system (by touching it, altering it, or being in the middle) with the intent to cause harm or make unauthorized changes.

Manomettere/Alterare

Definition: “interfere with (something) in order to cause damage or make unauthorized ^{changes} alterations”

ARE

Because tampering is a huge risk, engineers design sensitive systems using two defensive approaches:

- Tamper-Resistance: Creating physical or logical barriers that make it extremely difficult to alter the system.
- Tamper-Evident: If I cannot prevent a hacker from accessing the system, I ensure that any tampering is immediately visible. In the physical world, security seals (sigilli di sicurezza) are used (stickers that break if you try to open them); in the software world, encryption is used (e.g., hash functions or digital signatures) to instantly detect if even a single bit of a file has been altered.

composed of **hardware AND software**

physical tampering with a device's components, circuits, or logic for malicious purposes. Tampering requires that the attacker have physical access to the hardware.

- The hardware can be **tampered** by an attacker →
- Even the **physical security** of systems and devices (**e.g. to deny unauthorized access**) is hard to guarantee

How can you know if a hardware device has been tampered with? To ensure that a hardware device has not been tampered with, you must ensure the physical security of the device (since, if a hardware device has been tampered with, it is very difficult to detect).

Principle 2: on “Entities composing a system”



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HARDWARE

The “evil maid attack” is an example of unauthorized alteration of personal devices

composed of **hardware AND software**

- The hardware can be **tampered** by an attacker
- Even the **physical security** of systems and devices (**e.g. to deny unauthorized access**) is hard to guarantee

Principle 2: on “Entities composing a system”



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Possible to put a backdoor in a binary firmware without knowing the source code (it is easy to place it in the middle of the firmware)

To prevent unauthorized modifications, modern CPUs rely on hardware-based tamper-resistance, which cryptographically signs and verifies the firmware before execution.

HARDWARE

The Physical security of an HW must be guaranteed across the full hardware lifecycle, from initial purchase to final disposal (because if control were lost even for just a few seconds, it could be tampered). We do this because it is very difficult to detect if the HW it was tempered

It is an attack that exploits temporary physical access to compromise a device so that the attacker can control it remotely (by inserting a backdoor into the device's firmware).

The “**evil maid attack**” is an example of
unauthorized alteration of personal devices

Evil Maid Attack Demo

A “**nice demo**”: the time required
to flash the tampered firmware is
very limited

- The hardware can be **tampered** by an attacker
- Even the **physical security** of systems and devices (**e.g. to deny unauthorized access**) is **hard to guarantee**

Principle 2: on “Entities composing a system”



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- Lack of knowledge: The user or administrator simply does not know that a certain action is dangerous (e.g., reusing the same password for work and personal websites).
- Overload & Fatigue: A tired employee, overwhelmed by countless emails and deadlines, has a much lower attention span. It is in that moment of fatigue that they will click on the phishing email without checking it carefully.
- Bad design: If a security system is poorly designed, frustrating, and too complicated to use, the user will try to bypass it just to get their work done.
- Goodwill: Humans are inclined to help others. A hacker can exploit this “goodwill” by posing as a colleague in trouble who urgently needs a file or access to avoid losing a client.
- Lack of attention: A one-second distraction can lead to sending a confidential document to the wrong recipient.
- Stupidity (in both users and administrators): attitudes of total laziness or unjustifiable negligence. For example, a user who deliberately ignores antivirus security warnings, or a system administrator who leaves the main server password as “admin/admin” or “123456.”

HUMANWARE

No matter how secure the hardware or software is, humans remain the primary target for attackers.

- The “**human in the loop**” is often the **weakest link in the system**

(e.g. **social engineering**)

Attackers exploit human psychology (fear, urgency, trust) rather than technical flaws to bypass defenses (e.g., Phishing).

- For **many reasons**: lack of knowledge, overload, fatigue, bad design, goodwill, lack of attention... stupidity (*in both **users** and **system administrators***)

Principle 2: on “Entities

Social engineering is widely recognized as the most common and effective method for cyberattacks in the history of cybersecurity, making the “human element” the weakest link in security.

Definition: “the use of ^{inganno} deception to manipulate ^{to push him/her} individuals into divulging confidential or ^(that goes against cybersecurity policies) personal information that may be used for fraudulent purposes” (e.g. phishing emails)

- The “**human in the loop**” is often the **weakest link** in the system (e.g. **social engineering**)
- For **many reasons**: lack of knowledge, ^{sovraccarico (può essere indotto dall'attaccante)} overload, fatigue, bad ^{of tools} design, goodwill, lack of attention... stupidity (in both **users** and **system administrators**)

In such cases, people tend to take shortcuts by not following cybersecurity policies, which makes the system vulnerable

Principle 2: on “Entities composing a system”



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HUMANWARE

Correct way to mitigate this problem is through procedures (properly designed) to identify if I'm talking with a colleague or attacker

- The “**human in the loop**”
(e.g. **social engineering**)

buona volontà
Is “**goodwill**” a **problem**?

(example: faking to be a colleague and ask for confidential info)

Yes, when abused by attackers

- For **many reasons**: lack of knowledge, overload, fatigue, bad design, **goodwill**, lack of attention... stupidity (in both **users** and **system administrators**)

We are stupid when we don't understand something

Principle 2: on “Entities composing a system”



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Operational Security (OPSEC) is the process of protecting sensitive information by looking at your own operations through the eyes of an attacker.

SECURITY AS A PROCESS

- Cybersecurity isn't a product
Securing a system is a **[never ending] process** and **not a product**
(Cybersecurity is a day-by-day investment)
- **Reasonably secure system** requires **an endless work** on the system and **education**
- The system that today is “secure”, tomorrow will not be. **Everyday new faults are discovered**
secure
- The **lack of updates** leads to **fragile** and **insecure** systems

Principle 2: on “Entities composing a system”



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EVEN UPDATING IS NOT EASY...

- Are the updates **available** for all your devices? (e.g. **smartphone**)
- In other words, **is your devices supported? Until when?**



Principle 2: on “Entities composing a system”

System Administrators focus on Updates (when to do it, how, etc..)

EVEN UPDATING IS NOT EASY...

● Are the updates **available** for all your devices? (e.g. **smartphone**)

● In other words, **is your devices supported? Until when?**

if the device is no longer supported, the device is vulnerable if they discover a vulnerability and you cannot update it

→ The tech guy need to define the UPDATE POLICY, based on Company Context

● If the updates are available...

Installing everything blindly is dangerous. Updates can break existing system dependencies, cause software conflicts, or introduce new bugs (Choose which part of the update to do).

○ **what update to apply?** (*All of them... are you sure of that?*)

○ Timing is a trade-off. Updating live systems (in production) causes downtime. Scheduling updates during off-peak hours (e.g., night) minimizes impact but requires on-call engineering support (a rotation where engineers are tasked with monitoring systems and mitigating production incidents outside of normal business hours).

○ **when do you apply updates?** (*During the **night**? In “**production**”?*)

○ You must analyze the patch note. Security-critical hotfixes require immediate deployment, while feature-rich updates might be delayed to prioritize system stability.

○ **why?** (*Is that for **security** or to add **new functionalities**?*)

○ **how?** (*What is the best way to apply the update?* ^(Automatic Update) **Unattended?**)

Choosing the update method:

- Unattended (Automated): High efficiency but risks catastrophic, unmonitored failures.
- The Staging Strategy: Patches should be tested in an isolated staging environment before being pushed to production.

Principle 2: on “Entities composing a system”



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Since updating involves risks, a system administrator cannot blindly apply patches as soon as they are released. They must weigh the pros and cons:

- The benefits: What security vulnerabilities am I fixing? What risks do I face if I don't update?
- The costs: How much does it cost to shut down the machines for the update? What is the financial cost if something goes wrong and the system stops working?

EVEN UPDATING IS NOT EASY...

- The **updating process** is always a dangerous operation
 - for example: fixing bugs can introduce new bugs, instability, side-effects... due to complexity of the system we are dealing with

- What we need is a **costs/benefits evaluation**

- that is **costly** (e.g. time consuming)

Analyzing the impact of an update requires time and financial resources. Companies often need to test updates in a staging environment that mirrors the production environment before deploying them to production. This process requires hours of work and qualified personnel.

to take decision on a specific update

- it is often based on **partial information** (i.e. often there is a **lack of detailed information on the updates content**)

Without knowing exactly what has been changed in the code, those responsible for managing the system are forced to make critical decisions based on incomplete information, increasing uncertainty and the risks involved in the operation.

Principle 3: on “Security = Knowledge”



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KNOWLEDGE OF THE SYSTEM

Trust is even more
important than Knowledge

- Real security is based on **deep knowledge** (of the system to be protected)
- What are the **effects** of this principle **on software**?
 - **Open Systems** vs. **Closed Systems**
 - **Open Source** vs. **Closed Source**

Principle 3: on “Security = Knowledge”



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KNOWLEDGE OF THE SYSTEM

- Real security is based on **deep** knowledge (not just protected)

The **specifications** of the system are
OPEN / CLOSED

be

- What are the **effects** of this principle on software?
 - **Open Systems** vs. **Closed Systems**
 - **Open Source** vs. **Closed Source**

Prof. Opinion

I really prefer **OPEN** specifications (e.g. for interoperability) ^{with other system} but this does not guarantees that the system is secure (e.g. due to design or implementation faults)

- Real security (based on deep knowledge) is not protected)
- What are the effects of this principle on software?

- **Open Systems vs. Closed Systems**
- **Open Source vs. Closed Source**

Knowledge"

OF THE SYSTEM

If they have specs open/closed, doesn't guarantee security by definition

The specifications of the system are

OPEN / CLOSED

be

This means that the system's operating rules are a business secret. They are the exclusive property of a company that keeps them under cover. No one outside that company knows exactly how the data is processed or packaged for transmission. If you want to create something that interacts with that system, you either can't do it, or you have to pay a huge amount of money for a license.

This means that the details of how the system is designed, how it works, and how it communicates are public, free, detailed, and accessible to anyone (often standardized by international organizations). Anyone can read the technical manuals and build software or hardware that is fully compatible with that system, without having to ask permission or pay royalties to anyone.

Principle 3: on “Security = Knowledge”



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KNOWLEDGE OF THE SYSTEM

- Real security is based on **deep knowledge** (of the system to be protected)
- What are the **effects** of this principle **on software**?
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Principle 3: on “Security = Knowledge”



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KNOWLEDGE OF THE SYSTEM

- Real security is based on **knowledge** (i.e. **source code** protected)
- What are the **effects** of this principle on **software**?
 - **Open Systems** vs. **Closed Systems**
 - **Open Source** vs. **Closed Source**

The **source code** (i.e. **computer instructions**) used by the programmers to implement the system are

OPEN / CLOSED

Knowledge"

I really prefer **OPEN** Source (e.g. it can be *inspected, verified, validated* etc.) but this does not guarantees that the system is secure

GE OF THE SYSTEM

The source code (i.e. computer instructions) used by the programmers to implement the system are

OPEN / **CLOSED**

doesn't guarantee security by definition

- Real security is based on **distributed** protected
- What are the **effects** of this principle on software?
 - **Open systems** vs. **Closed Systems**
 - **Open Source** vs. **Closed Source**

- Closed Source (Security through Obscurity): Keeping source code secret under the assumption that attackers can't exploit what they can't see.
- Open Source (The Power of Crowdsourcing): Making source code public to maximize system knowledge.



I really prefer **OPEN** Source (e.g. it can be *inspected, verified, validated* etc.) but this does **not** guarantees that the system is secure

Even if it is theoretically possible (for me) to check all the code in a system (e.g. an open source Operating System like GNU/Linux), this is not practically possible (since it is made of millions of lines of source code)

Even if i have the source code of the program, it is difficult to know if the binary version of it contains a backdoor or malicious code that is different from the source code that is public available

- Real security based on protected
- What are the effects of this principle on software?
 - Open systems vs. Closed Systems
 - **Open Source** vs. Closed Source

Principle 3: on “Security = Knowledge”



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Management of Trust is fundamental in Cybersecurity and based on a lot of different stuff (example: choosing the browser based on history of vuln. fixing)

KNOWLEDGE OF THE SYSTEM

baseline of CySec (define the allies and enemies: This means determining which components are authorized to access sensitive data and which ones must be blocked or constantly monitored.)

- Real security is based on knowledge (of the system to be protected)

- What are the **effects**

- **Open Systems**

- **Open Source v**

In most cases it is a matter of

Security is based on an understanding (knowledge) of the system we want to protect, but

TRUST

is more important than knowledge (because it is very difficult to know/verify/check every single detail of the system)

Do you prefer to trust a
community (i.e. Open Source) or
a company (i.e. Close Source)?

Product of a Manufacture is based on Software supply chain (built on libraries, services, etc.): if something bad happens the responsibility is on the Manufacture (this is why they need to certificate them to build trust on Manufacture's product)

In the future, ensuring supply chain security will shift from a model of total verification to a scalable trust model. Organizations will increasingly rely on internal and external certifications to validate their cybersecurity levels and ensure compliance (conformità) for their customers.

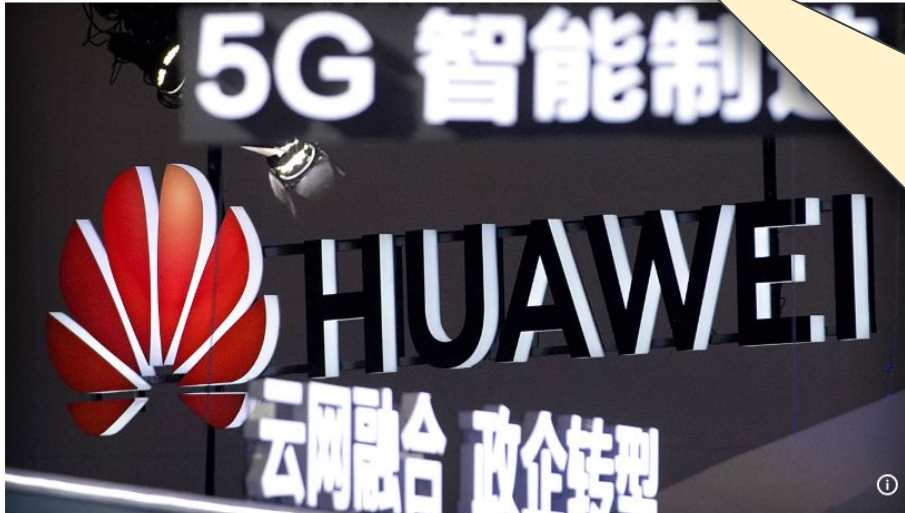
Nobody, now, create SW that is totally developed and/or designed in house (we use a lot of library, sw made by others, etc.): possible problem of the supply chain (example: (supply-chain attack) possible backdoor in our program that came from library developed by others)

Principle 3: on “Security = Knowledge”



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Eleven EU countries took 5G security measures to ban Huawei, ZTE



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By [Cynthia Kroet](#)

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because they cannot go in deep on the software of these devices to check about vulnerabilities like backdoors (they restricted because they don't have trust on some companies: possible that China gather info through it (not verified))

«Eleven countries [...] have used legal powers to impose **restrictions** on telecom suppliers that are considered high-risk, such as Huawei and ZTE, for 5G network infrastructure, a European Commission spokesperson told Euronews»

Principle 3: on “Security = Knowledge”



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KNOWLEDGE OF THE SYSTEM

- **Users' education** is a form of **knowledge**
- **Investing in education** is good but **not that simple**:
 - education is **costly**
 - **perception of security**: [lack of] importance of cybersecurity
 - [wrong] **habits** (i.e. “we have always done it this way”)
 - **ideological resistance** (i.e. “information anarchism”) (example: don't use passwords)
 - **cost of security** procedures (vs. “productive investments”)

An investment in Cybersecurity does not generate a profit; it merely serves to prevent a loss. For a manager focused on short-term numbers, it is always difficult to choose to spend money on security rather than on productivity. So, because compliance is mandatory, regulations are the primary leverage used to force business leaders to budget for cybersecurity over short-term productivity.

It is important but a few cares about it (increasing in these days)

○ **perception of security**: [lack of] importance of cybersecurity

○ [wrong] **habits** (i.e. “we have always done it this way”)

Some users reject security rules for ideological reasons.

○ **ideological resistance** (i.e. “information anarchism”) (example: don't use passwords)

○ **cost of security** procedures (vs. “productive investments”)

Addendum: “Security of a Cryptosystem”



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Principle that is important for the Professor

KERCKHOFFS'S PRINCIPLE



the security of a cryptographic system shouldn't rely on the secrecy of the algorithm

“A cryptosystem should be secure even if everything about the system, except the key, is public knowledge” (1883)

Addendum: “Security of a Cryptosystem”



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KERCKHOFFS'S PRINCIPLE



Algo that we use for encrypting the data, must be public (the only part that must be secret is the key, a small piece of information)

“A **cryptosystem** should be secure **even if everything about the system, except the key, is public knowledge**” (1883)

It is against Security through obscurity

It is a mechanism (i.e. hardware or
algorithm software) used for encrypting information

with the aim to preserve confidentiality

(encryption to preserve this property)

Addendum: “Security of a Cryptosystem”



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KERCKHOFFS'S PRINCIPLE

- “A **cryptosystem** should be secure **even if everything about the system, except the key, is public knowledge**” (1883)

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KERCKHOFFS'S PRINCIPLE

- “A **cryptosystem** should be secure **even if everything about the system, except the key, is public knowledge**” (1883)

SHANNON'S REFORMULATION



“The **enemy knows the system**”

not every secret of the system, but the design of the system

It is much better to base the security on the selection of the proper algo, not on the secrecy of it (maintaining private the key)

Addendum: “Security of a Cryptosystem”



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SECURITY THROUGH OBSCURITY

Wrong principle: against
Kerckhoffs' principle

- Means relying on the design or implementation secrecy as the main method of providing security

It is very difficult to maintain during the years

Why Obscurity Fails in Modern Cybersecurity:

- The Single Point of Failure: In Obscurity, if the design/algorithm is leaked or reverse-engineered, security drops to zero. Fixing it requires rewriting the entire system.
- The Key-Based Solution (Kerckhoffs): If a secret key is compromised, the algorithm remains secure. You only need to change the key, a fast, zero-cost operation.
- The Power of Peer Review: Public algorithms are verified by global cryptographic communities. Flaws are found and patched by friendly researchers before attackers can exploit them.
- The Reality of Reverse Engineering: Modern analysis tools ensure that an attacker will eventually discover how a system works. Secrecy is a temporary delay, not a security barrier.
- Overconfidence: This often leads developers to ignore real vulnerabilities because they think, “No one will ever find them anyway.”

Addendum: “Security of a Cryptosystem”



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SECURITY THROUGH OBSCURITY

- Means relying on the **design** or **implementation secrecy** as the main method of providing security
- The **security through obscurity** is **WRONG**

Addendum: “Security of a Cryptosystem”



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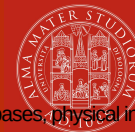
The **security through obscurity** is **WRONG**

- Does this means that I must **publish all the details about my system internals?**

When you develop a new application, there are two separate path that you can choose to adopt encryption:

- 1) Use standards (Good Approach: study papers and choose the perfect one)
- 2) Create a new algo for encryption (Stupid Approach: will do something with a lot of vulnerabilities and errors, if you are not an expert)

Addendum: “Security of a Cryptosystem”



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Before deploying any security control, you must define a Threat Model:

1. **WHAT** are you securing? (Asset Identification) You cannot protect everything equally. You must catalog your high-value assets (e.g., source code, customer PII, financial databases, physical infrastructure) and map their boundaries.

2. **WHY** are you securing it and against **WHO**? (Threat Analysis) Understand the specific risks and adversary motives. Are you preventing data theft (Confidentiality), unauthorized changes (Integrity), or system downtime (Availability)?

So, a security mechanism is useless without a clear model. The threat model defines the mechanism, ensuring that the defense matches the actual risk and asset value: first, create a threat model, and only then decide which technology or defense to implement (the mechanism). If you skip this step, you risk spending millions to fortify a secondary door while leaving the front door wide open. Risk should guide your choice of tool, never the opposite.

The security through obscurity is **WRONG**

- Does this mean that I must **publish all the details about my system internals?**
- **NO**, it would be very stupid! →

The security must not be based on secrecy, but this doesn't mean that you have to publish all details (you need to increase the cost for attacker to breach the system or gather info). Before disclosing information, evaluate whether it is truly necessary. Excessive disclosure fuels OSINT (Open Source Intelligence), allowing attackers to gather the data needed to refine and launch a more effective attack.

When we build up a security mechanism, we are totally losing common sense and avoid only cosmetic security mechanism: the security mechanism is useful if it is possible to enforce it (users will try to bypass it)



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Attributions and Notes



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- These slides are partially based on the work of Prof. **Renzo Davoli**
- **Ludovico Gardenghi** has worked on a previous version of these slides
- I wish to **thank all of them**, **any errors that remain are my sole responsibility**